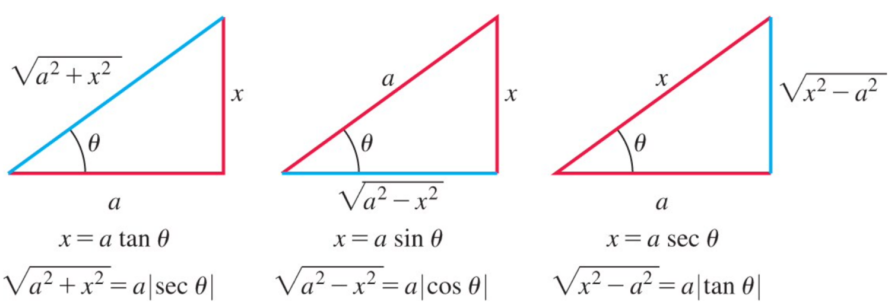
8.4 Trigonometric Substitution

❖ For integrals involving $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$ and $\sqrt{x^2-a^2}$.

❖ 3 basic trigonometric substitutions are $x=a\sin\theta$, $x=a\tan\theta$ and $x=a\sec\theta$.

Form	Substitution
$\sqrt{a^2-x^2}$	$x=a\sin\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$ (Use $1-\sin^2\theta=\cos^2\theta$ to eliminate the square root.)
$\sqrt{a^2+x^2}$	$x=a\tan\theta$, $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$ (Use $1+\tan^2\theta=\sec^2\theta$)
$\sqrt{x^2-a^2}$	$x=a\sec\theta$, $0\le\theta\le\frac{\pi}{2}$ (Use $\sec^2\theta-1=\tan^2\theta$)



Reference triangles for the three basic substitutions, identifying the sides labeled x and a for each substitution.

With $x=a\tan\theta$, $a^2+x^2=a^2+(a\tan\theta)^2=a^2(1+\tan^2\theta)=a^2\sec^2\theta$.

With $x=a\sin\theta$, $a^2-x^2=a^2-(a\sin\theta)^2=a^2(1+\sin^2\theta)=a^2\cos^2\theta$.

With $x=a\sec\theta$, $x^2-a^2=(a\sec\theta)^2-a^2=a^2(\sec^2\theta-1)=a^2\tan^2\theta$.

Any substitution we use in an integration need to be reversible so that we can change back to the original variable afterward. From Section 7.6, the functions in these substitutions have inverses only for selected values of θ. For reversibility,

$x=a\ \tan\ \theta$ requires $\theta=\arctan\left(\frac{x}{a}\right)$ with $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$,

$x=a\ \sin\ \theta$ requires $\theta=\arcsin\left(\frac{x}{a}\right)$ with $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$,

$x=a\ \sec\ \theta$ requires $\theta=\operatorname{arcsec}\left(\frac{x}{a}\right)$ with $\begin{cases}0\le\theta<\frac{\pi}{2} \text{ if } \frac{x}{a}\ge 1,\\ \frac{\pi}{2}<\theta\le\pi \text{ if } \frac{x}{a}\le -1.\end{cases}$

To simplify calculations with the substitution $x=a\sec\theta$, we will restrict its use to integrals in which $x/a\ge 1$. This will place θ in $[0,\pi/2]$ and make $\tan\theta\ge 0$. We will then have ,
With $\sqrt{x^2-a^2}=\sqrt{a^2\tan^2\theta}=|a\tan\theta|=|a\tan\theta|$, provided $a>0$.

Remark:

(1) For $\sqrt{a^2-(bx)^2}$, substitute $x=\frac{a}{b}\sin\theta$.

(2) For $\sqrt{a^2+(bx)^2}$, substitute $x=\frac{a}{b}\tan\theta$.

(1) For $\sqrt{(bx)^2-a^2}$, substitute $x=\frac{a}{b}\sec\theta$.

Procedure for a trigonometric substitution

1. Write down the substitution for x, calculate the differential dx, and specify the selected values of θ for the substitution.
2. Substitute the trigonometric expression and the calculated differential into the integrand, and then simplify the results algebraically.
3. Evaluate the trigonometric integral, keeping in mind the restrictions on the angle θ for reversibility.
4. Draw an appropriate reference triangle to reverse the substitution in the integration result and convert it back to the original variable x.

Example 1: Evaluate $\int \frac{dx}{\sqrt{4+x^2}}$.

Solution:

Let $x = 2\tan\theta,\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

Then $dx = 2\sec^2\theta d\theta$ and

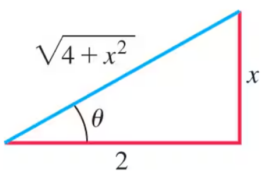
$$\sqrt{4+x^2} = \sqrt{4+(2\tan\theta)^2} = 2\sqrt{1+\tan^2\theta} = 2\sqrt{\sec^2\theta}$$

$$= 2|\sec\theta| = 2\sec\theta \qquad \left(\because \sec\theta > 0 \text{ for } -\tfrac{\pi}{2} < \theta < \tfrac{\pi}{2}\right).$$

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\theta d\theta}{2\sec\theta} = \int \sec\theta\, d\theta$$

$$= \ln|\sec\theta + \tan\theta| + C$$

$$= \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C.$$



$\tan\theta = \frac{x}{2} \Rightarrow \sec\theta = \frac{\sqrt{4+x^2}}{2}$

Example 2:
Find an expression for the inverse hyperbolic sine function in terms of the natural logarithm.

Solution:
Following the same procedure as in Example 1 with $x = a\tan\theta,\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$, we find that

$$\int \frac{dx}{\sqrt{a+x^2}} = \int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C = \ln\left|\frac{\sqrt{a^2+x^2}}{a} + \frac{x}{a}\right| + C.$$

From Table 8.1 (in Section 8.1), $\sinh^{-1}\left(\frac{x}{a}\right)$ is also an antiderivative of $\frac{1}{\sqrt{a+x^2}}$, so the two antiderivatives differ by a constant, giving

$$\sinh^{-1}\left(\frac{x}{a}\right) = \ln\left|\frac{\sqrt{a^2+x^2}}{a} + \frac{x}{a}\right| + C. \quad \text{-------- (1)}$$

Setting $x = 0$ in the equation (1), we have $0 = \ln 1 + C$, so $C = 0$.

Since $\sqrt{a^2+x^2} > |x|$, we conclude that $\sqrt{a^2+x^2} + ax > 0$, and therefore

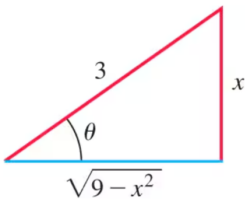
$$\sinh^{-1}\left(\frac{x}{a}\right) = \ln\left(\frac{\sqrt{a^2+x^2}}{a} + \frac{x}{a}\right).$$

Example 3: Evaluate $\int \frac{x^2 dx}{\sqrt{9-x^2}}$.

Solution:
Let $x = 3\sin\theta,\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = 3\cos\theta d\theta$ and

$$\sqrt{9-x^2} = \sqrt{9-(3\sin\theta)^2} = 3\sqrt{1-\sin^2\theta} = 3\sqrt{\cos^2\theta}.$$

$$= 3|\cos\theta| = 3\cos\theta \quad \left(\because \cos\theta > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}\right).$$



$$\int \frac{x^2 dx}{\sqrt{9-x^2}} = \int \frac{(3\sin\theta)^2 3\cos\theta d\theta}{3\cos\theta} = 9\int \sin^2\theta d\theta = 9\int \frac{1-\cos 2\theta}{2} d\theta = \frac{9}{2}\left(\theta - \frac{\sin 2\theta}{2}\right) + C$$

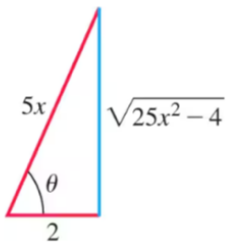
$$= \frac{9}{2}\left(\theta - \frac{2\sin\theta\cos\theta}{2}\right) + C = \frac{9}{2}\left(\arcsin\left(\frac{x}{3}\right) - \frac{x}{3}\left(\frac{\sqrt{9-x^2}}{3}\right)\right) + C. \qquad \sin\theta = \frac{x}{3} \Rightarrow \cos\theta = \frac{\sqrt{9-x^2}}{3}$$

Example 4: Evaluate $\int \frac{dx}{\sqrt{25x^2-4}},\ x > \frac{2}{5}$.

Solution:
Let $x = \frac{2}{5}\sec\theta,\ 0 < \theta < \frac{\pi}{2}$. Then $dx = \frac{2}{5}\sec\theta\tan\theta d\theta$ and

$$\sqrt{25x^2-4} = \sqrt{25\left(\frac{2}{5}\sec\theta\right)^2 - 4} = 2\sqrt{\sec^2\theta - 1} = 2\sqrt{\tan^2\theta}$$

$$= 2|\tan\theta| = 2\tan\theta \qquad \left(\because \tan\theta > 0 \text{ for } 0 < \theta < \frac{\pi}{2}\right).$$



$$\int \frac{dx}{\sqrt{25x^2-4}} = \int \frac{\frac{2}{5}\sec\theta\tan\theta d\theta}{2\tan\theta} = \frac{1}{5}\int \sec\theta d\theta$$

$$= \frac{1}{5}\ln|\sec\theta + \tan\theta| + C = \frac{1}{5}\left|\frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2}\right| + C. \qquad \sec\theta = \frac{5x}{2} \Rightarrow \tan\theta = \frac{\sqrt{25x^2-4}}{2}$$

Example 5: Use a suitable substitution to evaluate

(1) $\int \dfrac{dx}{\sqrt{16+25x^2}}$

(2) $\int \sqrt{9-4x^2}\,dx$

(3) $\int \dfrac{12dx}{x^2\sqrt{x^2-36}}$

(4). $\int_0^{\sqrt{3}} \dfrac{2x^2dx}{(4-x^2)^{3/2}}$

Solution:

(1) Let $x=\frac{4\tan\theta}{5}$, $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$. Then $dx=\frac{4}{5}\sec^2\theta d\theta$ and

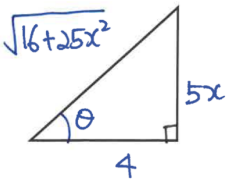
$$\sqrt{16+25x^2}=\sqrt{16+25\left(\frac{4\tan\theta}{5}\right)^2}=4\sqrt{1+\tan^2\theta}=4\sqrt{\sec^2\theta}$$

$$=4|\sec\theta|=4\sec\theta \qquad \left(\because \sec\theta>0 \text{ for } -\frac{\pi}{2}<\theta<\frac{\pi}{2}\right).$$

$$\int\frac{1}{\sqrt{16+25x^2}}dx=\int\frac{1}{4\sec\theta}\left(\frac{4}{5}\sec^2\theta\,d\theta\right)=\frac{1}{5}\int\sec\theta\,d\theta=\frac{1}{5}\ln|\sec\theta+\tan\theta|+C$$

$$=\frac{1}{5}\ln\left|\frac{\sqrt{16+25x^2}}{4}+\frac{5x}{4}\right|+C$$

$$=\frac{1}{20}\ln\left(\sqrt{16+25x^2}+5x\right)+C_1.$$



$\tan\theta=\frac{5x}{4}\Rightarrow\sec\theta=\frac{\sqrt{16+25x^2}}{4}$

(2) Let $x=\dfrac{3\sin\theta}{2}, -\dfrac{\pi}{2}<\theta<\dfrac{\pi}{2}$. Then $dx=\dfrac{3}{2}\cos\theta d\theta$ and

$$\sqrt{9-4x^2}=\sqrt{9-4\left(\frac{3\sin\theta}{2}\right)^2}=3\sqrt{1-\sin^2\theta}=3\sqrt{\cos^2\theta}$$

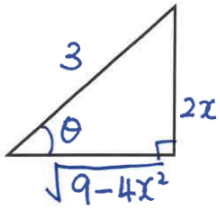
$$=3|\cos\theta|=3\cos\theta \qquad \left(\because \cos\theta>0 \text{ for } -\frac{\pi}{2}<\theta<\frac{\pi}{2}\right).$$

$$\int\sqrt{9-4x^2}\,dx=\int(3\cos\theta)\left(\frac{3}{2}\cos\theta\,d\theta\right)=\frac{9}{2}\int\cos^2\theta\,d\theta$$

$$=\frac{9}{2}\int\frac{\cos2\theta+1}{2}d\theta=\frac{9}{4}\left(\frac{\sin2\theta}{2}+\theta\right)+C$$

$$=\frac{9}{4}(\sin\theta\cos\theta+\theta)+C$$

$$=\frac{9}{4}\left[\frac{2x}{3}\left(\frac{\sqrt{9-4x^2}}{3}\right)+\sin^{-1}\left(\frac{2x}{3}\right)\right]+C=\frac{x\sqrt{9-4x^2}}{2}+\frac{9}{4}\sin^{-1}\left(\frac{2x}{3}\right)+C.$$



$\sin\theta=\frac{2x}{3}\Rightarrow\cos\theta=\frac{\sqrt{9-4x^2}}{3}$

(3) Let $x = 6\sec\theta, 0 < \theta < \frac{\pi}{2}$. Then $dx = 6\sec\theta\tan\theta d\theta$ and

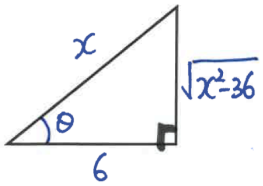
$\sqrt{x^2 - 36} = \sqrt{36\sec^2\theta - 36} = 6\sqrt{\sec^2\theta - 1} = 6\sqrt{\tan^2\theta}$

$= 6|\tan\theta| = 6\tan\theta \qquad \left(\because \quad \tan\theta > 0 \text{ for } 0 < \theta < \frac{\pi}{2}\right).$

$$\int \frac{12dx}{x^2\sqrt{x^2-36}} = \int \frac{12}{(6\sec\theta)^2(6\tan\theta)}(6\sec\theta\tan\theta\, d\theta)$$

$$= \frac{1}{3}\int \frac{1}{\sec\theta} d\theta = \frac{1}{3}\int \cos\theta\, d\theta$$

$$= \frac{1}{3}\sin\theta + C = \frac{\sqrt{x^2-36}}{3x} + C.$$



$\sec\theta = \frac{x}{6} \Rightarrow \sin\theta = \frac{\sqrt{x^2-36}}{x}$

(4) Let $x = 2\sin\theta$. Then $dx = 2\cos\theta d\theta$.

When $x = 0$, $\theta = 0$; $x = \sqrt{3}$, $\theta = \frac{\pi}{3}$.

$\sqrt{4-x^2} = \sqrt{4-(2\sin\theta)^2} = 2\sqrt{\cos^2\theta} = 2|\cos\theta| = 2\cos\theta \quad \left(\because \quad \cos\theta > 0 \text{ for } 0 < \theta < \frac{\pi}{3}\right)$

$$\int_0^{\sqrt{3}} \frac{2x^2dx}{(4-x^2)^{3/2}} = \int_0^{\pi/3} \frac{2(4\sin^2\theta)}{(2\cos\theta)^3} 2\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/3} \tan^2\theta\, d\theta = 2\int_0^{\pi/3} (\sec^2\theta - 1) d\theta$$

$$= 2\left[\tan\theta - \theta\right]_0^{\pi/3} = 2\left[\sqrt{3} - \frac{\pi}{3}\right].$$